

MIT 6.087 — Lecture 2_2: Operators & Expressions

Practical Programming in C • Arithmetic, Bitwise, Logic & Type Conversions

1. Arithmetic Operators (+, -, *, /, %)

- Integer Division vs Float Division: $3 / 2$ results in 1 (truncated integer division), whereas $3.0 / 2$ results in 1.5.
- Modulus (%): Yields integer remainder (e.g. $7 \% 4$ is 3). Operands must be integers.

2. Relational & Equality Operators

- Operators: $>$, $>=$, $<$, $<=$, $==$, $!=$
- Gotcha: The $==$ equality test operator is distinct from the $=$ assignment operator.

3. Logical Operators & Short-Circuit Evaluation

- $\&\&$ (Logical AND), $\|\|$ (Logical OR), $!$ (Logical NOT)
- Short-circuit: $(3 == 3) \|\| (c = getchar()) == 'y';$ -> Second expression is skipped!

Bitwise Operators, Precedence & Type Conversions

1. Bitwise Operators

- & (Bitwise AND), | (Bitwise OR), ^ (Bitwise XOR), << (Left Shift), >> (Right Shift)
- Example: 0x77 & 0x03 evaluates to 0x03; 0x01 << 4 evaluates to 0x10 (16).

2. Increment & Decrement (++ / --)

- Postfix (x++): Value of x is used in expression first, then incremented.
- Prefix (++x): Value of x is incremented first, then used in expression.

3. Operator Precedence Hierarchy

- Highest: ++, --, (cast), sizeof
- High: *, /, % followed by +, -
- Medium: Relational (<, <=, >, >=), Equality (==, !=)
- Lower: Logical (&&, ||), Conditional (?:)
- Lowest: Assignment operators (=, +=, -=, *=, /=, %=)